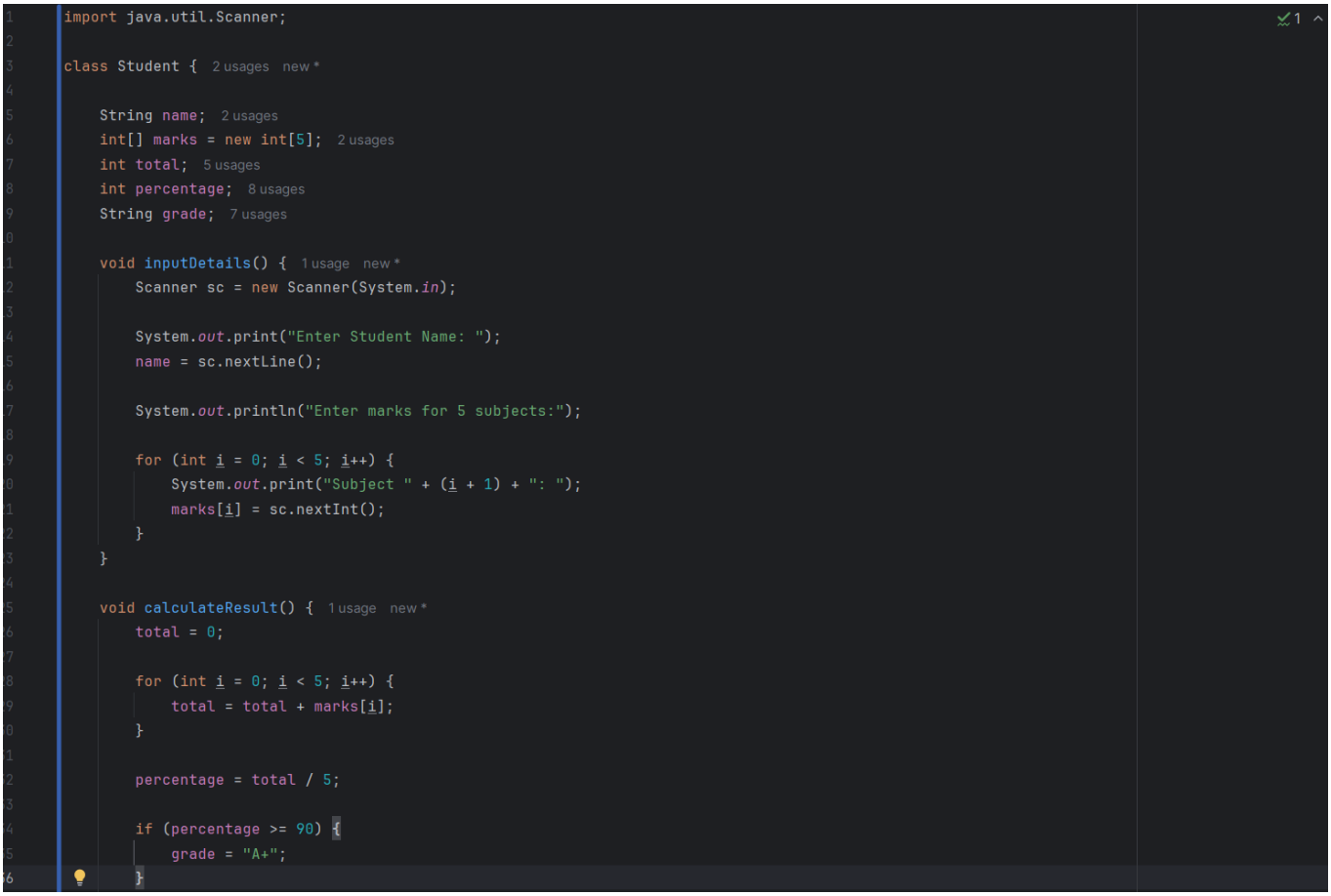
Java

Case study –1

Student Examination Result

1 code



```java
import java.util.Scanner;

class Student {  2 usages  new *

    String name;  2 usages
    int[] marks = new int[5];  2 usages
    int total;  5 usages
    int percentage;  8 usages
    String grade;  7 usages

    void inputDetails() {  1 usage  new *
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Student Name: ");
        name = sc.nextLine();

        System.out.println("Enter marks for 5 subjects:");

        for (int i = 0; i < 5; i++) {
            System.out.print("Subject " + (i + 1) + ": ");
            marks[i] = sc.nextInt();
        }
    }

    void calculateResult() {  1 usage  new *
        total = 0;

        for (int i = 0; i < 5; i++) {
            total = total + marks[i];
        }

        percentage = total / 5;

        if (percentage >= 90) {
            grade = "A+";
        }
```

```
        else if (percentage >= 80) {
            grade = "A";
        }
        else if (percentage >= 70) {
            grade = "B";
        }
        else if (percentage >= 60) {
            grade = "C";
        }
        else if (percentage >= 50) {
            grade = "D";
        }
        else {
            grade = "F";
        }
    }

    void displayResult() {  1 usage  new *

        System.out.println("\n----- STUDENT RESULT -----");
        System.out.println("Name: " + name);
        System.out.println("Total Marks: " + total + "/500");
        System.out.println("Percentage: " + percentage + "%");
        System.out.println("Grade: " + grade);

        if (percentage >= 50) {
            System.out.println("Result: PASS");
        }
        else {
            System.out.println("Result: FAIL");
        }
    }
}
```

```java
public class casestudy1 {  Dinesh *
    public static void main(String[] args) {  new *
        Student s1 = new Student();

        s1.inputDetails();
        s1.calculateResult();
        s1.displayResult();
    }
}
```

2 output

```
Enter Student Name: Dinesh
Enter marks for 5 subjects:
Subject 1: 98
Subject 2: 97
Subject 3: 96
Subject 4: 95
Subject 5: 94

----- STUDENT RESULT -----
Name: Dinesh
Total Marks: 480/500
Percentage: 96%
Grade: A+
Result: PASS

Process finished with exit code 0
```